Minimax Allocation of Limited Human Review Across Protected Groups

T

Abstract

A team auditing an agent has a fixed number of reviews and may not know which user group bears the harm. We derive the best worst-group detection probability for a finite audit with equal review costs. A continuous calculation is exact because the detection curves are concave and their fractional allocations admit budget-preserving randomization. The resulting plan satisfies integer review floors on every draw. Independent enumeration verifies 128 two-group instances, including cases where randomization improves on every fixed allocation. We extend the result to imperfect review and derive the exact guarantee when miss events can be dependent. A bound on pairwise joint detections gives a sharp intermediate curve without specifying the full dependence structure.

Summary and scope

Question. How should a service allocate a fixed number of human reviews when it does not know which group is affected?

Finding. Concave interpolation gives the exact worst-group detection optimum. Dependent rounding implements its allocation while satisfying all integer review floors and the total slot budget on every draw.

Evidence and scope. The result is proved and checked on synthetic queues with independent enumeration. It gives a way to reserve attention for sparse groups. The guarantee assumes disjoint groups, equal review costs and stated minimum harm counts. The imperfect-review extension also needs a lower sensitivity bound and a declared model for dependence between misses. Pairwise bounds can tighten the guarantee without assuming independence.

1 One budget and several places to look

A service can audit a scheduling or eligibility agent only after human reviewers inspect the affected cases. A group with few reports may be small, poorly represented, or simply difficult to reach. Proportional sampling can miss harm concentrated in such a group. A separate minimum review count can guarantee that each group receives attention, but equal counts do not imply equal chances of finding harm.

Partition a fixed queue into disjoint strata $g = 1, \dots, G$ with n_g cases. Every review costs one slot and observes a binary harm label. The unknown affected stratum contains at least $h_g \geq 1$ harmed cases if it is affected. Their positions are unknown, and the reviewer has no reliable case-level signal about them before inspection. There are B slots and integer floors a_g, with $0 \leq a_g \leq n_g$

and $\sum_g a_g \le B$. Allocate without replacing a reviewed case. The objective is to maximize the lowest probability of detecting at least one harm when the affected stratum is chosen adversarially. Additional affected strata can only help the detection probability of a fixed randomized allocation.

For a uniform sample of b cases in stratum g, the detection probability at its minimum harm count is

$$f_g(b) = 1 - \frac{\binom{n_g - h_g}{b}}{\binom{n_g}{b}},$$

where the numerator is zero if too few unharmed cases remain. Let $\bar{f}_g$ interpolate f_g linearly between adjacent integers. The result below relies on the equal slot cost and on that interpolation, rather than on replacing finite sampling by independent draws with replacement.

2 An exact continuous allocation

Theorem 1. *The best worst-stratum detection probability over all randomized audit designs is*

$$\max_{\substack{a_g \le x_g \le n_g \\ \sum_g x_g \le B}} \min_g \bar{f}_g(x_g).$$

For any maximizing vector x, there is a randomized design whose integer counts satisfy every floor and the total budget in every realization. Its expected count in group g is x_g, and its worst-case detection probability in that group is $\bar{f}_g(x_g)$.

Proof. First consider the no-harm sequence of observations. For every fixed choice of the design's random coins, this sequence determines which cases it would inspect up to budget, including all adaptive decisions on that sequence. When exactly one stratum is harmed, the design detects harm if and only if this hypothetical path intersects its harmed set. Before the first hit all observations match the hypothetical path. Thus adaptations based solely on the binary review outcome cannot improve the event of the first detection beyond the induced random no-hit allocation.

For a given stratum, put its h_g harmed cases uniformly among its n_g cases for the upper-bound argument. With b_g no-hit path reviews in that stratum, the detection probability averaged over placements is $f_g(b_g)$. The worst placement therefore has detection probability at most $\mathbb{E} f_g(b_g)$. Take $x_g = \mathbb{E} b_g$. The increments of f_g are nonnegative and nonincreasing. In fact if $q_b = 1 - f_g(b) > 0$, then the next increment is $q_b h_g/(n_g - b)$. The ratio of successive positive increments is

$$\frac{n_g - h_g - b}{n_g - b - 1} \le 1$$

whenever both increments are defined and the denominator is positive. After certainty is reached the increments vanish. Hence $\bar{f}_g$ is concave and Jensen's inequality gives $\mathbb{E} f_g(b_g) \le \bar{f}_g(x_g)$. The vector x meets the continuous constraints, proving the upper bound.

For the converse, write $x_g = m_g + r_g$ with m_g integer and $0 \le r_g < 1$. Make consecutive half-open intervals of length r_g and draw one uniform shift U in $[0, 1)$. Round group g up when its interval contains some point of $U + \mathbb{Z}$. An interval of length less than one gets at most one such point, with probability exactly r_g. The total number of round-ups is at most $\lceil \sum_g r_g \rceil$. Since the budget is an integer, the rounded counts sum to at most B. The integer floors and caps are preserved. This is a dependent rounding construction [1] specialized to one shared budget.

Choose uniformly within each group using its rounded count. Every fixed harmed set of size h_g then has detection probability $\mathbb{E} f_g(b_g) = \bar{f}_g(x_g)$, because b_g takes only the two integers adjacent to x_g. Larger harmed sets have at least that probability. □

A target probability q is feasible if the sum of the smallest x_g with $\bar{f}_g(x_g) \geq q$ and $x_g \geq a_g$ is at most B. This gives a scalar search. Each inverse curve is piecewise linear. For rational finite data, finding the breakpoint interval that contains the target and interpolating gives the exact optimum.

3 Where randomization helps

With one slot and two groups of ten, one harmed case in the affected group, and no floors, every fixed allocation leaves one group unchecked. Choose each group with probability one half. Each has detection probability 1/20 in its worst placement. This is a guarantee averaged over the audit draw, so it does not guarantee that both groups are reviewed on that draw. Positive floors are the way to impose that separate requirement when the budget permits it.

For a larger example, take 80 cases with a minimum of eight harms in the first stratum, 20 cases with a minimum of one in the second, and ten review slots. The optimal mixed design has expected counts approximately 3.605 and 6.395. Its worst-stratum detection probability is 18121/56674, about 0.31974. The best fixed allocation gives 0.30. Splitting the slots equally gives 0.25 and proportional allocation gives 0.10. Here the smaller stratum receives more reviews because its stipulated harm density is lower.

The verifier compares the continuous result against independent two-group enumeration of all feasible allocations and their line-segment mixtures. In the plane, an optimum of the smaller coordinate over the convex hull is attained at a vertex or on an edge, so that oracle is exact. All 128 optima agree. Randomization strictly improves the fixed allocation in 30 cases, with a largest gap of 2/5. An independent interval calculation checks budgets, floors, marginals and achieved detection on 128 additional three-group cases. These synthetic counts describe the fixed experiment design.

4 Imperfect reviews and shared misses

An inspection can miss a real harm. Suppose each harmed case in group g has a latent binary mark indicating whether a review would detect it. The mark law is fixed with the harm placement before the audit's sampling coins. Clean cases never raise a hit. The latent marks can represent a shared blind spot in a rubric or independent reviewer misses. We assume a common lower bound $s_g \in [0, 1]$ on each harmed case's marginal detection probability conditional on the fixed placement.

Theorem 2. *If dependence between marks is unrestricted, replace $f_g(b)$ in the minimax allocation theorem by $s_g f_g(b)$. If marks are independent, replace it by*

$$f_g^{\mathrm{ind}}(b) = \sum_{m=0}^{h_g} \binom{h_g}{m} s_g^m (1 - s_g)^{h_g - m} \left[1 - \frac{\binom{n_g - m}{b}}{\binom{n_g}{b}} \right].$$

In either regime, interpolate the replacement curve and use the same budget-preserving rounding rule.

Proof. With a uniform sample of size b, condition on which cases it contains. If it contains a harmed case, pick any one of them using a fixed order on the cases. Its mark is positive with probability at least s_g. Independence of the sampling coins from the mark law makes the bound valid for this conditional sample. Thus detection probability is at least $s_g f_g(b)$. For the matching upper bound, give the h_g harmed cases a single common Bernoulli mark of mean s_g. Use a uniform harm placement as in the first theorem. The no-hit path analysis gives exactly $s_g f_g(b)$ after averaging over placements and the common mark.

For independent marks, the number of detectable harmed cases stochastically dominates a binomial with parameters h_g, s_g. Conditional on the actual number m, uniform review finds a mark with probability $1 - \binom{n_g - m}{b} / \binom{n_g}{b}$. This gives the displayed lower bound. For the upper bound, take exactly h_g uniformly placed harms with independent marks of mean s_g. Conditional on their number, detectable cases are uniform among subsets of that size. The no-hit path then has the displayed detection curve. Both replacement curves are concave after interpolation. The first is a nonnegative multiple of f_g, and the second is a mixture of finite-population curves. Jensen's inequality and the original rounding proof apply. □

The shared-miss model caps guaranteed detection at s_g even after reviewing the entire group. With independent marks the cap is $1-(1-s_g)^{h_g}$. The scalar allocation search therefore starts with the smallest group cap as its upper endpoint. In a fixture with group sizes 16 and 10, minimum harm counts five and three, sensitivities 1/2 and 3/4, and ten review slots, the optimal guaranteed detection is 0.4882 under arbitrary dependence and 0.6744 under independence. The first group receives about 7.328 expected reviews in the dependence-robust design and 6.119 in the independent design.

The verifier checks 1,980 joint-law and slot-count combinations on one five-case queue. Explicit mark and sample enumeration checks 100 independent curves. In 96 seeded two-group fixtures, an allocation oracle confirms both optima, and independence strictly raises the optimum in 41. These are finite checks of the stated models. Estimating sensitivity and testing for common misses require data from the proposed review process.

4.1 What pairwise detection evidence can buy

Suppose each harmed case is detectable with probability at least $s > 0$, and each pair is jointly detectable with probability at most u. These are bounds for a fixed harm placement, with the joint mark law chosen before sampling. Let M be the number of positive marks among h harmed cases. The bounds imply $\mathbb{E}M \geq hs$ and $\mathbb{E}[M(M-1)] \leq h(h-1)u$.

Write $F_b(m)$ for the detection probability under b uniform reviews when exactly m of n cases have a positive mark. Define

$$\mu = hs, \qquad t = \min\{h, 1 + (h-1)u/s\}.$$

Let Z_* have mean t and support on the two adjacent integers around t, using a point mass when t is integral. For $m \geq 1$ set

$$p_m = \frac{\mu}{m} \Pr(Z_* = m), \qquad p_0 = 1 - \sum_{m=1}^{h} p_m.$$

Theorem 3. *The marginal and pairwise bounds are jointly feasible exactly when $p_0 \geq 0$. In that case the sharp detection guarantee for every sample size b is $\sum_m p_m F_b(m)$. The same distribution (p_m) attains all these minima, and the resulting curve can replace f_g in the allocation theorem.*

Proof. To minimize detection, it suffices to consider $\mathbb{E}M = \mu$. For a law with larger mean, independently replace all its marks with zero often enough to reach μ. This preserves the pairwise caps and can only lower detection. Define Z by weighting each positive count by its size, so $\Pr(Z = m) = m \Pr(M = m)/\mu$. The second-moment bound gives $\mathbb{E}Z \leq t$. For any function F with $F(0) = 0$, $\mathbb{E}F(M) = \mu \mathbb{E}[F(Z)/Z]$.

The sequence $F_b(m)/m$ is decreasing and discretely convex. To see this for $b > 0$, count a hit by giving each selected marked case a share equal to the reciprocal of the total marks selected. Conditioning on inclusion of one marked case yields

$$\frac{F_b(m)}{m} = \frac{b}{n}\,\mathbb{E}\left[\frac{1}{1+K_m}\right],$$

where K_m counts marked cases among $b-1$ other sampled cases when $m-1$ of the $n-1$ remaining cases are marked. Couple the counts for successive m by marking new cases in a fixed order while retaining the same random sample. The reciprocal decreases. For two new marks, the expected second difference equals the expected mixed difference from adding each mark separately, since interchanging their positions preserves the sample law. It is zero unless both new marks are sampled. If both are sampled, it is a nonnegative second difference of $k \mapsto 1/(1+k)$. For $b = 0$ the sequence is identically zero.

The linear interpolant of this decreasing convex sequence and Jensen's inequality give the minimum at the mean t, with equality under Z_*. The same argument for $1/m$ shows that Z_* minimizes $\mu\mathbb{E}(1/Z)$, the probability of a positive count. Thus p_0 must be nonnegative whenever the bounds are feasible. If it is nonnegative, sample a count from (p_m) and mark a uniformly chosen subset of that size. The marginal probabilities are s and the pairwise probabilities satisfy the cap. This realizes the claimed curve. With more than h harmed cases the lower bound still holds by restricting attention to any h of them. The curve is a mixture of concave finite-population curves, and the uniform placement and no-hit argument from the first theorem gives the matching minimax allocation bound. □

With three harmed cases, $s = 1/2$ and $u = 1/4$ give $t = 2$. The least favorable law has two marks with probability $3/4$ and zero marks otherwise. A full review therefore guarantees detection with probability $3/4$. The corresponding guarantees are $1/2$ under unrestricted dependence and $7/8$ under independence. The pair cap here is the same as the joint probability under independence, yet even exact pairwise probabilities leave room for a shared blind spot across all three cases.

The exact verifier enumerated the vertices of the count moment polytope for 1,296 panels. It found 274 infeasible panels and matched 6,682 sample-size minima on the feasible ones. An audit can use a joint-detection bound derived from adjudicated paired cases if the design justifies transferring that bound to the queue. The formula supplies the detection guarantee conditional on that bound. It does not estimate it from unaudited cases.

5 When the model needs to change

Unequal review costs can invalidate the interpolation. For example, let there be one case per group, review costs two and one, and budget one. A fractional calculation with expected counts $(1/3, 1/3)$ satisfies the expected budget but no feasible realization can inspect the first group. Its worst-group detection probability is zero. In that setting a mixed strategy must be supported on actually feasible cost allocations. Overlapping group membership also changes the feasible set because one review can count toward several floors.

A prospective audit should preregister group definitions, actual slot costs, and the harm counts for which it reports power. The method can balance scarce review effort for an agent's decisions, while a reliable predictor, variable review duration, or false positive labels would change the optimization. When review sensitivity depends on assignments or on prior reviews, evaluate that process before adopting either mark law. Report the realized allocation, the audit design's detection probability, and any failed coverage floors as separate facts.

Review status. Independent manuscript review is pending.

References

[1] Rajiv Gandhi, Samir Khuller, Srinivasan Parthasarathy, and Aravind Srinivasan. Dependent Rounding and Its Applications to Approximation Algorithms. Journal of the ACM 53(3), 2006. `https://www.cs.umd.edu/~srin/PDF/2005/deprounding-dec05.pdf`.